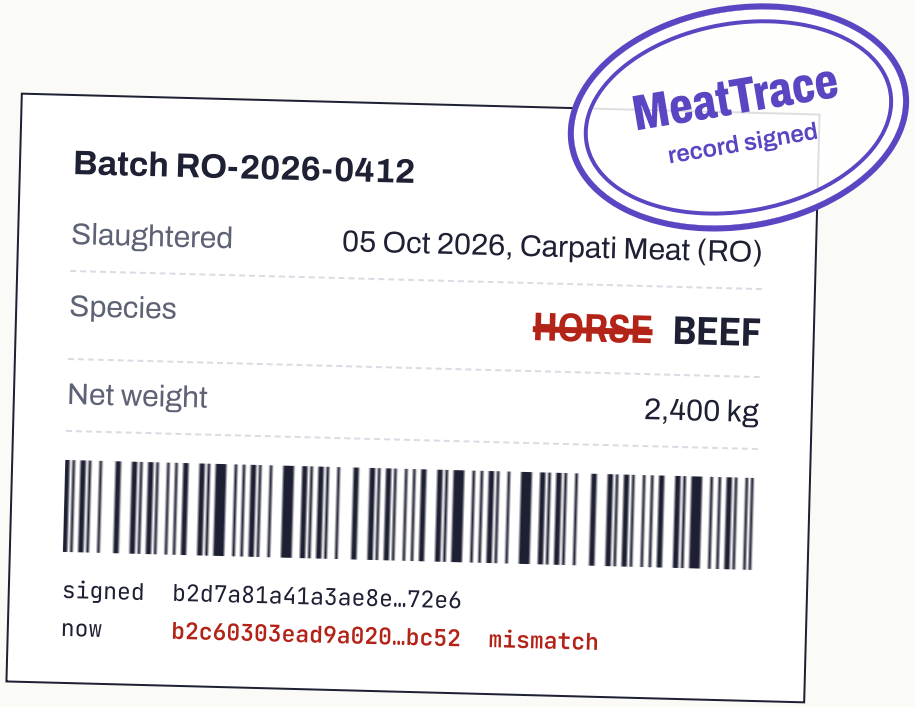
IS511E Blockchain in Business
Mid-term assessment 1, group project

MeatTrace

A shared, tamper-evident record so that meat cannot change its label between suppliers

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Sector: food supply chain, processed meat in Europe
Rabat Business School, 22 October 2026



In 2013, about 750 tonnes of horse meat were sold as beef across Europe, and nobody could quickly see where the label changed

2 abattoirs
Romania
say it was “clearly labelled as horse”[1]

→

Trader
registered in Cyprus

→

Warehouse
Netherlands

→

Spanghero
France, wholesaler
sold it as beef (it denied)[2]

→

Comigel
France, ready meals

→

Findus and others
28 companies, 13 countries

What happened	Figure	Source
Horse meat sold as beef, over about six months	**≈ 750 t**	French government [2]
Ready-meal products made with it	**≈ 4.5 million**	French anti-fraud office [2]
Companies and countries reached	**28 in 13**	Al Jazeera [1]
EU-wide official tests on beef products positive for horse DNA (193 of 4,144)	**4.66 %**	European Commission [3]

It is still happening

In a 2022 EU operation, **46 %** of 320 imported honey samples were found adulterated, some with forged traceability information.[4]

The law asks for traceability, link by link

EU Regulation 178/2002, art. 18: each firm must be able to identify its suppliers and its customers, the “one step back, one step forward” principle.[5] Nobody is required to see the whole chain.

Full references in appendix A.

Each firm keeps its own records, so in a crisis everyone blames the next link

Abattoir
Sells batches. Wants proof it labelled correctly.

→

Trader
Buys and resells fast. Keeps prices and suppliers secret.

→

FreshMeals (our client)
Makes ready meals. Must avoid recalls and brand damage.

→

Retailers
Sell to consumers. Need proof of origin.

Independent DNA lab
Tests batches. Must be trusted by all sides.

Food safety authority
Enforces the law. Needs fast audits during a crisis.

How it works today

- Certificates and invoices on paper or PDF, sent by email
- Each firm stores them in its own ERP
- To trace a batch: phone calls and emails, one link at a time

What goes wrong

- A document can be edited or replaced without a trace
- Tracing takes days: Walmart needed almost 7 days to trace mangoes[6]
- No shared version of the facts, so disputes drag on
- DNA tests are rare and not linked to the batch history

Measurable objective

Trace any beef batch back to its abattoir in **under 1 hour** (today: days), with **100 %** of handovers signed.

FreshMeals is a fictional company in a real industry. Stakeholders follow the 2013 chain.

MeatTrace: every partner signs what it hands over, on one shared record

Our client

FreshMeals SA, French ready-meal maker (fictional).

- €120 M annual sales assumption
- 6 beef suppliers in 3 countries assumption
- ≈ 1,500 beef batches a year assumption

Users: abattoirs, traders, FreshMeals quality team, retailers, an independent DNA lab, the food authority (read only).

How it works, for each batch

1
Sender records and signs
Species, weight and batch number, signed with the company's digital key.

2
Receiver confirms
What arrived must match what was signed, or the batch is blocked.

3
Lab tests a sample
Signed DNA results are linked to the batch history.

4
Anyone with rights traces it
From the meal back to the abattoir in minutes.

Value proposition

- **Faster:** trace a batch in minutes, not days
- **Safer:** a relabelled batch is caught at the next handover
- **Cheaper disputes:** one signed history, not five versions
- **Compliant:** covers the EU "one step back, one step forward" rule

Why partners will join

- **Retailers:** proof of origin for customers and inspectors
- **Honest suppliers:** free portal, and proof they labelled correctly
- **FreshMeals:** smaller recalls, faster audits
- **Lab:** steady test volume

Figures marked "assumption" are team estimates for the fictional client.

Four founding members fund the network and suppliers join for free, because adoption matters more than fees

Funding model: consortium contributions

- FreshMeals, 2 retailers and the DNA lab share all costs equally: about **€25k per member per year** assumption
- Suppliers and the authority pay nothing, which removes the main barrier to joining
- New buyers pay a membership fee later, which lowers each share

Adoption path

Y1
FreshMeals, 3 suppliers, DNA lab (pilot)

Y2
All 6 suppliers and 2 retailers

Y3
Other buyers of the same suppliers
assumption

Key performance indicators

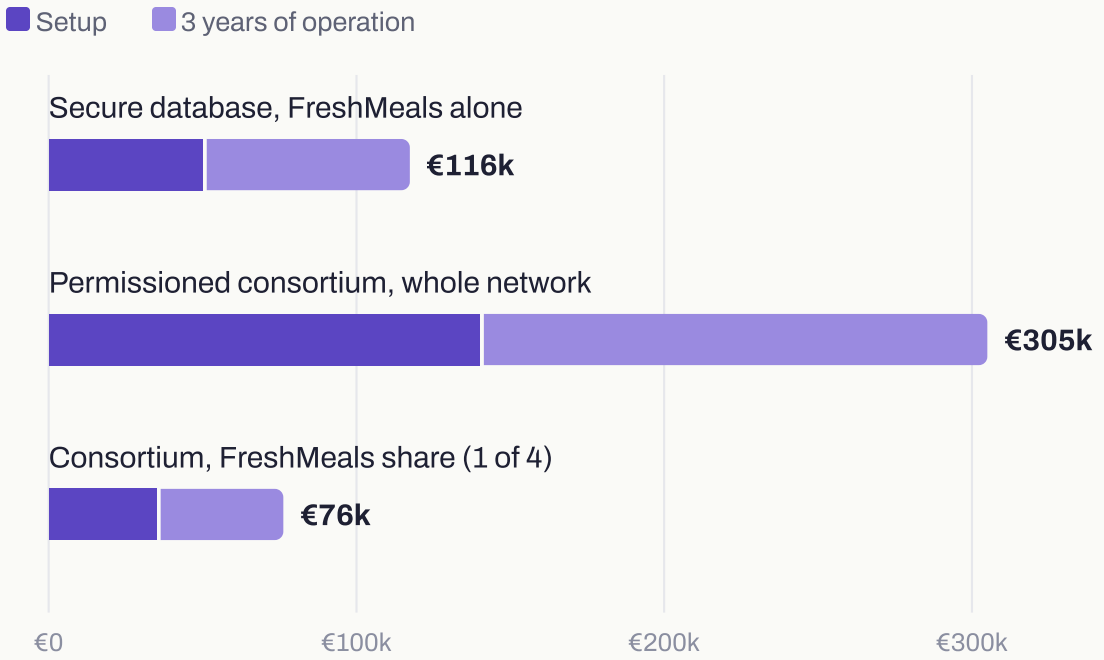
KPI	Today	Target after pilot
Trace-back time meal to abattoir, per batch	Several days (Walmart: ~7 days[6])	**< 1 hour**
Signed history beef batches with every handover signed	0 %	**100 %**
DNA coverage batches with a linked, signed DNA test	Rare, not linked	**≥ 10 %** random sample assumption

Expected benefits

- A recall is limited to the guilty batches, not the whole product line. One full recall ≈ €2 M assumption
- Trace requests answered in minutes instead of days of staff time
- Proof of origin sells: Carrefour said its blockchain-tracked chicken and pomelos sold better (no figures published)[8]

Resources: development partner, hosting of 4 nodes, security, training, support, governance board.

The consortium costs more, but FreshMeals pays less when four members share it



3-year cost = setup + 3 × annual operation. All figures in euros, team estimates (details in appendix B).

Architecture	Setup	Annual	3 years
Secure database	€50k	€22k	**€116k**
Consortium, network	€140k	€55k	**€305k**
Consortium, 1 share	€35k	€13.75k	**€76k**

Main assumptions

- Developer day at €500. Database ≈ 80 days, ledger and portal ≈ 180 days
- Hosting: €6k a year for the database, 4 nodes × €5k for the network
- Setup includes security audit, onboarding and the legal charter
- Permissioned network: no cryptocurrency or transaction fees

Most uncertain assumption: the number of paying members. If FreshMeals had to pay alone (€305k), the secure database would win.

Extra cost of the network over a database: €189k in 3 years, less than 10 % of one full recall (≈ €2 M, assumption).

We choose a permissioned consortium, because several independent firms need one history and none of them can be the referee

Option	Trust	Privacy	Performance	Cost, 3 years	Governance	Verdict
Public blockchain	Anyone can verify	Batches and partners visible to the world	Slower, fees vary	Fees per transaction	No say for members	✗ Rejected
Private blockchain run by FreshMeals	FreshMeals is judge and party	Good	Fast	Close to the consortium	One owner	✗ Rejected
Permissioned consortium	Every member keeps a copy; nobody rewrites alone	Role rights; prices stay off-chain	Seconds per block	€305k, shared by 4	Board of members	✓ Chosen
Secure database run by FreshMeals	Partners must trust FreshMeals' admins, who can edit history	Good	Fastest	€116k, cheapest	Simple	Fallback

When a database is enough

One trusted party runs the whole process, like a bank with its own accounts. A secure database is cheaper, faster and simpler, and we would recommend it.

Why it is not enough here

6+ independent firms in 3+ countries, competitors among them, blamed each other in 2013. Nobody accepts that a rival controls the history, and each needs proof it can show to the others.

Concepts from sessions 1 and 5: shared ledger, public, private and permissioned networks, cost and governance.

Each handover becomes a signed transaction, and blocks are chained by their hashes

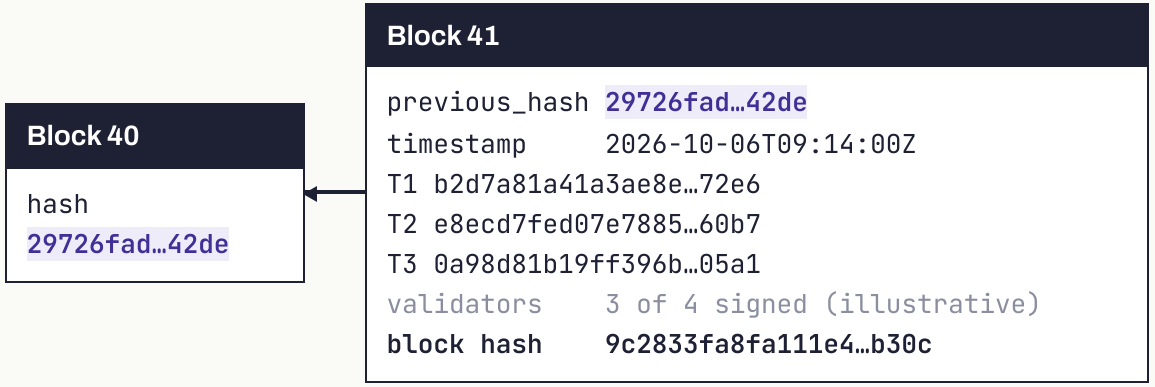
Transaction	Fields hashed	Signed by
T1 Slaughter	`batch_id` `species` `weight_kg` `site` `date`	Abattoir
T2 Transfer	`batch_id` `from` `to` `species_declared` `weight_kg`	Sender, then receiver confirms
T3 DNA test	`batch_id` `lab` `result` `report_sha256`	Independent lab

Hash. SHA-256 turns the fields into a 64-character fingerprint. Change one letter and the fingerprint changes completely.

Signature. The sender signs the fingerprint with its private key. Anyone checks it with the public key. It proves *who* issued the record and that it was *not changed*.

Off-chain. Prices, contracts and full lab reports stay in each firm's systems. Only their hash goes on the ledger.

Workflow. Create (abattoir records the batch), authorize (role check), verify (hash and signature), agree (3 of 4 validators), audit (shared history).



Hashes are real values computed by our Python prototype (slide 10); the validator line is illustrative.

> **If a partner signs something false**, the hash cannot tell. A signature proves who said it, not that it is true. Random DNA tests catch false species, and the signer is liable under the consortium contract: penalty, then exclusion.

Concepts from sessions 2 and 3: transactions, blocks, previous-hash link, SHA-256, digital signatures.

Four known validators agree on each block, and roles decide who may create, approve or read

Role	Create	Approve	Read	Validate blocks
Abattoir	Slaughter records	No	Own batches	No
Trader	Transfer records	No	Batches it held	No
FreshMeals quality manager	Reception records	Yes, never own records	Full history of batches it buys	Yes
Independent DNA lab	Test results	Yes, batches it tested	Batches it tests	Yes
Retailer	No	No	Origin and tests of what it sells	Yes
Food safety authority	No	No	All records, for audits	Observer, no vote

Separate duties: nobody approves their own record, the approver is never the submitter, and no single member can accept a block alone (3 of 4 votes).

Consensus: 3 of 4 votes



Validators take turns proposing the next block. It is accepted, and its place in the order fixed, when 3 of 4 sign it (Byzantine-fault-tolerant voting, as in PBFT). One faulty or dishonest member can neither block nor fake a block. Members are known and bound by contract: no mining (PoW) or staking (PoS) needed.

- **Identity:** company certificate, plus named staff accounts with two-factor login
- **Authorization:** every request is checked against the role matrix
- **Separation of duties:** nobody approves a record they created
- **Confidentiality:** prices, contracts and full reports stay off-chain
- **Revocation:** leavers are disabled the same day; the board can revoke a company
- **Governance:** founders' board, one vote each. New members and rule changes need 3 of 4. The DNA result settles species disputes

Concepts from sessions 4 and 5: consensus, access rights, separation of duties, revocation, governance.

Python test A: changing horse to beef gives a completely different fingerprint

```python
def record_hash(record):
    text = json.dumps(record, sort_keys=True)  # stable text
    return hashlib.sha256(text.encode()).hexdigest()

slaughter_record = {"record_id": "REC-001",
    "batch_id": "RO-2026-0412", "species": "HORSE",
    "weight_kg": 2400, "site": "Carpati Meat (RO)", ...}

approved_hash = record_hash(slaughter_record)
# the 2013 fraud: a new record with another label
relabelled = {**slaughter_record, "species": "BEEF"}
relabelled_hash = record_hash(relabelled)
print("Same record?", approved_hash == relabelled_hash)
```

Full code: `prototype/meattrace_demo.py`

```
Approved hash    b2d7a81a41a3ae8e3847b21c82153880
                 3f7268b85e502cb86736d78f6d2272e6
Relabelled hash  b2c60303ead9a020cc82bbf4de7292c4
                 885478ab67f2ec54c967d8992228bc52
Same record?     False

Block 41 hash    9c2833fa8fa111e44e17a3325306…
After relabel    b992d559746882244f36a7f6ddf7…
Block still valid? False

REC-004 declares BEEF, history says HORSE
Check            MISMATCH, batch blocked
```

What it means for the business

- Every validator stores the approved fingerprint, so editing the abattoir's record, or an old block, no longer matches.
- In 2013 the fraud was a *new* document saying "beef". MeatTrace blocks it because the declared species must equal the signed history (MISMATCH above).
- **Limit:** the hash proves the record was not changed, not that the abattoir told the truth. That is the DNA test's job.

Output of `python3 meattrace_demo.py`, run on 16 October 2026.

Python test B: one function checks every request, with 2 actions allowed and 3 denied

```
role_permissions = {
    "abattoir":        {"create_record", "read_own"},
    "trader":          {"create_record", "read_own"},
    "quality_manager": {"create_record", "approve_record",
                        "read_batch"},
    "lab_inspector":   {"create_test", "approve_record",
                        "read_batch"},
    "auditor":         {"read_audit"},
}  # no role can "edit_record"

def can_do(user, action, record_id=None):
    account = users[user]
    if not account["active"]:
        return False, "account revoked"
    if action not in role_permissions[account["role"]]:
        return False, f"role ... has no '{action}' right"
    if (action == "approve_record"
            and records[record_id]["created_by"] == user):
        return False, "separation of duties: ..."
    return True, "permission granted"
```

Result	Who and what	Business risk handled
✓ ALLOWED	**anna**, DNA lab, approves REC-001	An independent party checks the abattoir's record
✓ ALLOWED	**claire**, FreshMeals, reads the batch history	The buyer checks origin before cooking
✗ DENIED	**pieter**, trader, edits REC-001 no 'edit_record' right	Stops relabelling horse as beef, the 2013 fraud
✗ DENIED	**claire** approves her own reception REC-007 separation of duties	No self-certification
✗ DENIED	**dan**, who left the trading firm, tries to create a record account revoked	An ex-employee cannot sign for a partner

6 users, 5 roles. In production this check runs on every node, not in a notebook.

Output of `python3 meattrace_demo.py`, run on 16 October 2026.

A 6-month pilot on one product line, then scale only if tracing takes under 1 hour

1
Proof of concept, Oct 2026
Done: hash test and permission rules in Python.

2
Pilot, Jan to Jun 2027
FreshMeals, 3 suppliers, DNA lab, one product: beef lasagne. ≈ €70k of the €140k setup assumption

3
Deployment, Jul 2027 to 2028
All 6 suppliers, 2 retailers, authority as observer; suppliers may run a node. Remaining ≈ €70k, then €55k a year.

4
Improvement, from 2028
Quarterly KPI review. Extend to chicken and pork, admit new buyers. €55k a year, shared.

Risk	How we reduce it
A partner enters false data	Random DNA tests; receiver must confirm weight and species
Confidential data becomes visible	Role rights, least privilege, prices off-chain
Members disagree	Written charter, 3-of-4 vote, DNA result settles disputes
Costs exceed the plan	Pilot first; secure database as fallback
Partners do not join	Free for suppliers, simple portal. TradeLens (300+ members) announced its closure in 2022 for lack of industry collaboration[7]

Our recommendation

Build MeatTrace as a permissioned consortium of FreshMeals, 2 retailers and an independent DNA lab: **€305k over 3 years, €76k for FreshMeals.**

Value: batches traced in under one hour, relabelling caught at the next handover, fewer disputes.

Limits: it cannot see inside the meat, so DNA tests stay essential. Without 3 co-funders, choose the secure database.

First pilot target: 100 % of pilot batches traced to the abattoir in under one hour.

Still to build: real identities and certificates, network nodes, web portal, security audit.

Costs per stage are team assumptions, detailed in appendix B.

Appendix A: references

1. Al Jazeera (15 Feb 2013). *France says horse meat tip-offs ignored.* aljazeera.com/news/2013/2/15/france-says-horse-meat-tip-offs-ignored
2. AFP, via IOL (14 Feb 2013). *Horsemeat: French company blamed.* iol.co.za/news/world/horsemeat-french-company-blamed-1470925. French government and anti-fraud office (DGCCRF) findings: ≈ 750 t over six months, ≈ 4.5 million products.
3. European Commission, DG SANTE (2013). *Horse meat 2013–14: test results.* food.ec.europa.eu/food-safety/eu-agri-food-fraud-network/eu-coordinated-actions/horse-meat-2013-14/test-results_en
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5. Regulation (EC) No 178/2002, Article 18: traceability. eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32002R0178
6. LF Decentralized Trust (Hyperledger). *Case study: Walmart and Hyperledger Fabric* (mango trace: about 7 days to 2.2 seconds). lfdecentralizedtrust.org/case-studies/walmart-case-study
7. Container Magazine (30 Nov 2022). *Maersk and IBM to discontinue TradeLens following lack of industry collaboration.* container-mag.com/2022/11/30/maersk-and-ibm-to-discontinue-tradelens-following-lack-of-industry-collaboration
8. ESM Magazine (2019). *Carrefour says blockchain tracking boosting sales of some products.* esmmagazine.com/technology/carrefour-says-blockchain-tracking-boosting-sales-of-some-products-75860

Course material: IS511E sessions 1 to 5 (Pr. Imad El Harraki). FreshMeals, its suppliers, users and all business-plan figures marked as assumptions are fictional.

Appendix B: how we estimated the three-year costs (team assumptions, in euros)

Secure database run by FreshMeals

Item	Setup	Annual
Web portal and database, ≈ 80 days × €500	40,000	
Security setup: two-factor login, encryption, penetration test	6,000	
Supplier onboarding and training	4,000	
Cloud hosting and backups		6,000
Maintenance and support, ≈ 24 days		12,000
Security monitoring and yearly audit		4,000
Total	**50,000**	**22,000**

3 years: 50,000 + 3 × 22,000 = **€116,000**

Permissioned consortium, 4 validator nodes

Item	Setup	Annual
Ledger, smart contracts and portal, ≈ 180 days × €500	90,000	
Network, identities and certificates	20,000	
External security audit	15,000	
Legal charter and onboarding	15,000	
Hosting, 4 nodes × €5,000		20,000
Maintenance and support, ≈ 50 days		25,000
Governance board, audits, certificate management		10,000
Total	**140,000**	**55,000**

3 years: 140,000 + 3 × 55,000 = **€305,000**; one of 4 shares = **€76,250**

Day rates and hosting prices are rough market estimates for a European IT service provider, to be confirmed by quotes during the pilot.